

The deviations at a glance

Ratios, SI translation and correction quantities

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The deviations at a glance

Abstract

This document records all residual deviations of FFGFT in one place, each with the measured value and its uncertainty. It follows the order of the theory: first the ratios of the basic form in natural units with $\alpha = 1$, then the translation into SI. The ratios agree to within per mille to a few per cent; the measurement uncertainties are almost everywhere orders of magnitude smaller than the deviations and contribute only a minimal amount. Three correction quantities carry the residuals: the Higgs correction β_T , which already appeared in the earliest documents of 2025 as a small deviation from $\beta_T = 1$, the fractal correction $K_{\text{frak}} = 74/75$ of the SI translation, and the factor 10 in ν/E_p . For each one, the document shows which variants exist, how well they fit and which connections arise.

Status markers

- [B]** Algebraically proven.
- [K]** Numerically confirmed (check script, comparison with measured values).
- [S]** Assumption, open or speculative.
- [X]** Refuted or not sustainable.
- [Q]** Convention.

Measured values: CODATA 2022, PDG 2024, neutrino data as in Doc. 340 with the uncertainties from PDG 2024. The column “measurement error” gives the relative uncertainty of the comparison value; where it is comparable to the deviation, the deviation is also given in standard deviations. Check script: `pruef_387_abweichungen.py`.

1 How to read the deviations

FFGFT is first and foremost ratio-based (Docs. 384, 386). In the basic form, in natural units with $\alpha = 1$ and a redefined charge, there is only $\xi = 1/7500$ and chosen integers; ratios test the structure. Absolute values in SI need an anchor and a reference energy and test the conversion (A130). The deviations are therefore listed separately. A deviation much larger than the measurement error is a genuine residual of the theory, not a measurement problem; this holds for almost all entries.

2 Ratios of the basic form

Quantity	FFGFT	Deviation	Meas. error	Status
m_μ/m_e	$120\sqrt{3} = 207.85$	+0.52 %	2×10^{-8}	[K] , genuine residual
m_τ/m_μ	$\frac{125}{144}\xi^{-1/3} = 16.99$	+1.03 %	5×10^{-5}	[K] , genuine residual
Koide $Q - \frac{2}{3}$	0	-0.017 %	0.038ξ	0.4σ
v/E_p	$\xi^4/(5\pi)$	-0.23 %	1×10^{-5}	[K] , factor 10
ξ from m_h/v	$\beta_T = 0.977$	-2.3 %	1.8×10^{-3}	check [K]
$\Delta m_{\text{atm}}^2/\Delta m_{\text{sol}}^2$	$120/(11/3) = 32.7$	-1.4 %	2.6 %	0.5σ
$\sin^2 \theta_{13}$	$(25\xi)^{2/3} = 0.0223$	+1.4 %	3 %	0.4σ
M_W/M_Z	$\sqrt{7/9}$	+0.06 %	1.7×10^{-4}	3.8σ
m_h/M_Z	11/8	+0.15 %	9×10^{-4}	1.7σ
m_t/M_Z	$(11/8)^2$	-0.10 %	1.7×10^{-3}	0.6σ
λ_{CKM}	$\xi^{1/6} = 0.2260$	+0.46 %	3×10^{-3}	1.5σ

The lepton ratios are the sharpest tests of the basic form. Their measurement errors are 10^{-8} and 10^{-5} , the deviations 0.5 and 1 %; the residuals are thus entirely residuals of the ladder, and the measurement contributes nothing **[K]**. Their cause is open (R113). For the neutrinos, θ_{13} , m_t and λ_{CKM} the deviations lie within 1.5 standard deviations; there the measurement is the limiting factor. M_W from $\sin^2 \theta_W = 2/9$, at 3.8σ , is the only stated tension in the basic form. Koide agrees within the measurement precision of m_τ .

3 The translation into SI

Quantity	FFGFT	Deviation	Meas. error	Status
α^{-1} with 74/75	137.059	$+1.7 \times 10^{-4}$	1.5×10^{-10}	[K] , anchor
α^{-1} Galois	$3700/27 = 137.037$	$+7.6 \times 10^{-6}$	1.5×10^{-10}	[K]
$m_e m_\mu$ against 54 MeV ²	54	-1.6×10^{-4}	2×10^{-8}	[K]
m_e (chain, measured E_p)	$\frac{4}{15\pi} \xi^{11/2} E_p$	-1.32 %	1×10^{-5}	[K]
m_μ (chain)	$\frac{16}{25\pi} \xi^5 E_p$	-0.80 %	1×10^{-5}	[K]
m_τ (chain)	$\frac{3}{9\pi} \xi^{14/3} E_p$	+0.22 %	5×10^{-5}	[K]
ν from the ladder, bare	248.93 GeV	+1.10 %	10^{-6}	[K]
ν from the ladder with K	245.61 GeV	-0.25 %	10^{-6}	[K]
Δm_{atm}^2	$2.476 \times 10^{-3} \text{ eV}^2$	-1.0 %	1.1 %	0.9σ
Δm_{sol}^2	$7.565 \times 10^{-5} \text{ eV}^2$	+0.46 %	2.4 %	0.2σ

Here too the measurement errors, except for the neutrinos, are orders of magnitude smaller than the deviations. Two residuals stand out. The residual of α , 1.7×10^{-4} , is exactly the distance of the measured masses from the Galois value: $m_e m_\mu = 53.991 \text{ MeV}^2$ lies 1.6×10^{-4} below $54 = |\text{GF}(3)^*| \cdot |\text{GF}(27)|$, and with 54, $\alpha^{-1} = 3700/27$ agrees to 7.6 ppm **[K]**. The residual of the electron mass in the chain, -1.32 %, equals $K_{\text{frak}} - 1 = -1.33 \%$ to within 1.6×10^{-4} **[K]**; this does not hold for the muon and the tau.

4 The correction quantities

β_T : the Higgs correction from the earliest documents

The earliest documents of March and April 2025 set two quantities to 1: $\alpha = 1$ and $\beta_T = 1$, on the grounds that both are conventions in natural units and that their SI values, $\alpha \approx 1/137$ and $\beta \approx 0.008$, arise only through the translation ("The consistency of $\alpha = 1$ and $\beta = 1$ ", version history). $\beta_T = 1$ with $r_0 = \xi \ell_P$ gave the Higgs formula $\xi = \lambda_h^2 \nu^2 / (16\pi^3 m_h^2)$; the document gave $r_0 \approx \ell_P/7519$ instead of $\ell_P/7500$, i.e. already then a small deviation from $\beta_T = 1$ (Doc. 385). Computed today,

$$\beta_T = \frac{\xi_{\text{EFT}}}{\xi} = \frac{m_h^2}{64\pi^3 \nu^2 \xi} = 0.977 \quad \mathbf{[K]}, \quad (1)$$

with a measurement error of 0.18 % from m_h . The deviation from $\beta_T = 1$ is therefore a genuine residual, 13 times larger than the measurement uncertainty. It depends on which ν is inserted:

Input for ν	ν	β_T
measured (from G_F)	246.22 GeV	0.977
chain $\xi^4 E_p / (5\pi)$	245.65 GeV	0.982
ladder with m_e and K_{frak}	245.61 GeV	0.982
ladder with $m_{e\tau}$ bare	248.93 GeV	0.956

With FFGFT's own values for ν , β_T moves closer to 1 but stays at -1.8 %; with the bare ν it moves away. The Higgs check remains a check of ξ to within a few per cent **[K]**. There is no derivation of the residual **[S]**. Numerically, $\beta_T = 0.9772$ lies close to $K_{\text{frak}}^{\sqrt{3}} = 0.9770$; with a measurement error of 0.18 % this is not a robust statement.

K_{frak} : the correction of the SI translation

In the basic form there is no correction factor; the bridge reads $\xi E_0^2 = 1$ (Doc. 386). In the SI translation, α requires the factor $K_\alpha = \xi m_e m_\mu \alpha^{-1} = 0.98650$. The variants compared:

Variant	K	α^{-1}	Galois	ν (ladder)	Turns
$1 - 100\xi = 74/75$	0.986667	$+1.7 \times 10^{-4}$	+7.6 ppm	−0.25 %	75.0
$(1 - \xi)^{100} \approx e^{-100\xi}$	0.986754	$+2.6 \times 10^{-4}$	+96 ppm	−0.24 %	75.5
$1/(1 + 100\xi)$	0.986842	$+3.5 \times 10^{-4}$	+190 ppm	−0.23 %	76.0
$1 - \psi'(75)$ (Doc. 381)	0.986577	$+7.8 \times 10^{-5}$	−83 ppm	−0.26 %	74.5
without correction	1	+1.4 %	+1.4 %	+1.10 %	–

Only 74/75 closes the winding after a whole number of turns and matches the Galois form to 7.6 ppm **[K]**. $1 - \psi'(75)$ halves the α residual but loses the whole number of turns and worsens the Galois form. The multiplicative and the inverse forms fit worse everywhere than the additive one; regarding the open form (R139), the α sector thus favours $1 - 100\xi$. No variant acts on the lepton ratios; for ν and G they differ only by 10^{-4} .

The factor 10 in ν/E_P

$\nu/E_P = \xi^4/(5\pi)$ with $5\pi = (\pi/2) \cdot 10$ agrees to −0.23 % at a measurement error of 10^{-5} . The factor 10 is fitted to the measured value: in the working state of February 2026, $m_P/(f^4 \cdot \pi/2)$ gave 2467 GeV instead of 246 GeV, and the gap of 10.02 was closed with 1/10. Two readings were examined:

- Rationalised gravitation ($8\pi G = 1$) with the reduced Planck mass would give $2\sqrt{8\pi} = 10.03$ instead of 10. Docs. 374 and 376, however, show that the appropriate reference energy is the unreduced Planck energy; the reduced reading would be a different physical assumption. This reading does not hold **[X]**.
- Measured versus bare ν : if the bare $\nu = 248.93$ GeV of the ladder is inserted instead of the measured $\nu = 246.22$ GeV, the formula requires the denominator 15.501. This is $5\pi K_{\text{frak}} = 15.4985$ or $\pi^3/2 = 15.5031$, each to within 1.5×10^{-4} . The factor would then be $10 K_{\text{frak}} \approx \pi^2$ instead of 10. The two cannot be distinguished at this level **[S]**.

The second reading fits the separation of basic form and SI: the factor 10 would then be a mix-up of the bare value of the ladder with the SI comparison value of ν . It is not a derivation.

On notation: here f is the base winding number $f = 1/\xi = 7500$, not a frequency. In natural units f is only another reading of the duality $T \cdot m = 1$: $f = t_P/t_0 = \Lambda_{T0}/E_P$ with $t_0 = \xi t_P$ and $\Lambda_{T0} = E_P/\xi$, and $\xi E_0^2 = 1$ gives $f = E_0^2$. As a dimensionless ratio, f cannot be set to 1 with justification (Doc. A135).

5 Connections

The residuals do not stand unconnected side by side:

1. **α and Galois.** The α residual of 1.7×10^{-4} is the distance of $m_e m_\mu$ from 54 MeV². With the Galois value, $K_{\text{frak}} = 74/75$ stays exact and the residual lies on the side of the masses **[K]**.
2. **Electron and K_{frak} .** The electron residual of the chain is $K_{\text{frak}} - 1$ to within 1.6×10^{-4} . The chain, however, contains the factor 10; as long as that is open, this agreement is not robust **[S]**.
3. **Factor 10 and K_{frak} .** With the bare ν , the factor 10 becomes $10 K_{\text{frak}}$; the same correction that carries E_0 and α would then also stand in ν/E_P **[S]**.

4. β_T . The Higgs correction, at -2.3% , is the largest and oldest. With FFGFT's own values for ν it drops to -1.8% ; it shows no robust connection with K_{frak} **[S]**.

6 Measurement precision

Input	relative uncertainty	effect
α, m_e	$1.5 \times 10^{-10}, 3 \times 10^{-10}$	negligible
m_μ, M_Z	$2 \times 10^{-8}, 2 \times 10^{-5}$	negligible
G (hence E_p)	$2 \times 10^{-5} (1 \times 10^{-5})$	negligible
m_τ	5×10^{-5}	negligible except for Koide
m_h	9×10^{-4}	β_T to 0.18%
m_t, M_W	$1.7 \times 10^{-3}, 1.7 \times 10^{-4}$	limiting
neutrino data, $\theta_{13}, \lambda_{\text{CKM}}$	1 to 3 %	limiting

For the lepton masses, α , ν/E_p and the SI chain the measurement errors contribute only a minimal amount; the residuals there are residuals of the theory. For m_h, m_t, M_W , the neutrinos and the mixing angles the measurement is the limiting factor, and the deviations are given there in standard deviations.

7 Conclusion

The ratios of the basic form agree to within per mille to a few per cent, and where the measurement is precise, the residuals are genuine residuals of the theory. Three correction quantities carry them. β_T is the oldest: the earliest documents already set $\beta_T = 1$ alongside $\alpha = 1$ and saw a small deviation; today it is 0.977 , and 0.982 with FFGFT's own ν . $K_{\text{frak}} = 74/75$ is the correction of the SI translation; among the variants it is the only one with a whole number of turns, and it matches the Galois form to 7.6 ppm. The factor 10 in ν/E_p is fitted; with the bare ν of the ladder it becomes $10 K_{\text{frak}} \approx \pi^2$. Open are the cause of the lepton residuals, the residual of β_T and the origin of the factor in ν/E_p .

Check script: `2/python/Dok387_Skripte/pruef_387_abweichungen.py` — 27/27.